HYDROLOGICAL PROCESSES
Hydrol. Process. **28**, 62–73 (2014)
Published online 17 October 2012 in Wiley Online Library
(wileyonlinelibrary.com) DOI: 10.1002/hyp.9548

An automated algorithm for river ice monitoring over the Susquehanna River using the MODIS data

Naira Chaouch,[1]* Marouane Temimi,[1] Peter Romanov,[1] Reggina Cabrera,[2] George McKillop[2] and Reza Khanbilvardi[1]

[1] *NOAA-CREST Institute, The City University of New York, New York, NY, USA*
[2] *National Oceanic and Atmospheric Administration–National Weather Service, Bohemia, NY, USA*

Abstract:

Reliable and prompt information on river ice condition and extent is needed to make accurate hydrological forecasts to predict ice jams breakups and issue timely flood warnings. This study presents a technique to detect and monitor river ice using observations from the MODIS instrument onboard the Terra satellite. The technique incorporates a threshold-based decision tree image classification algorithm to process MODIS data and to determine the extent of ice. To differentiate between ice-covered and ice-free pixels within the riverbed, the algorithm combines observations in the visible and near-infrared spectral bands. The developed technique presents the core of the MODIS-based river ice mapping system, which has been developed to support National Oceanic and Atmospheric Administration NWS's operations. The system has been tested over the Susquehanna River in northeastern USA, where ice jam events leading to spring floods are a frequent occurrence. The automated algorithm generates three products: daily ice maps, weekly composite ice maps and running cloud-free composite ice maps. The performance of the system was evaluated over nine winter seasons. The analysis of the derived products has revealed their good agreement with the aerial photography and with *in situ* observations-based ice charts. The probability of ice detection determined from the comparison of the product with the high-resolution Landsat imagery was equal to 91%. A consistent inverse relationship was found between the river discharge and the ice extent. The correlation between the discharge and the ice extent as determined from the weekly composite product reached 0.75. The developed CREST River Ice Observation System has been implemented at National Oceanic and Atmospheric Administration–Cooperative Remote Sensing Science and Technology Center as an operational Web tool allowing end users and forecasters to assess ice conditions on the river. Copyright © 2012 John Wiley & Sons, Ltd.

KEY WORDS River ice; MODIS data; image classification

Received 5 January 2012; Accepted 3 September 2012

INTRODUCTION

Ice jams on rivers in the Northern Hemisphere mid and high latitudes present one of the major factors causing early spring floods. In the United States, the annual estimated cost of damages caused by ice related hydrologic events is around $120 million (White *et al.*, 2007). The availability of accurate and detailed information on the ice conditions and distribution on rivers helps to better assess the flood risk, issue timely flood warnings and take proper preventive or mitigation measures. The river ice onset and breakup are fine-scale processes that may occur fast (Hicks and Beltaos, 2008). Air surveys and *in situ* observations are the main traditional ways to collect information on the river ice. However, such observations are either local or rare and in many cases are not sufficient to provide comprehensive information on the state of the river ice. In the USA, information on the state of the river ice is available from reports issued by the US Geological Survey (USGS) Water Resources Division and from USGS stream-gauging station data. Information on the ice jams is collected by the US Army Cold Regions Research and Engineering Laboratory (CRREL) and summarized in the Ice Jam Database (Ice Engineering Group, 1999). The use of satellite data to complement *in situ* data and air surveys appears as an attractive way to improve the river ice monitoring.

Owing to detailed spatial and frequent temporal coverage, satellite images have been used in the river monitoring for more than three decades (Dey *et al.*, 1977). Microwave observations, particularly active microwave measurements with radars, have been largely used for the ice mapping. The main advantage of microwave observations is in the ability of the microwave signal to penetrate through clouds. Coarse spatial resolution, however, particularly in the case of passive microwave measurements, limits their efficacy to study the river hydraulics (Temimi *et al.*, 2005; Brakenridge *et al.*, 2007). Satellite-based active microwave measurements, on the other hand, provide information on the distribution of the river ice at higher spatial resolution (Seidou *et al.*, 2006; Gherboudj *et al.*, 2007; Chokmani *et al.*, 2008; Leconte *et al.*, 2009). However, the main drawback of active microwave sensors such as Radarsat 1 and 2 is their

*Correspondence to: Naira Chaouch, National Oceanic and Atmospheric Administration–Cooperative Remote Sensing Science and Technology Center, The City University of New York, New York, NY, USA.
E-mail: nchaouch@ccny.cuny.edu

Copyright © 2012 John Wiley & Sons, Ltd.

narrow swath and hence long revisit time which reduces the value of active microwave data for operational applications.

Optical imaging sensors onboard meteorological sun-synchronous satellites offer frequent revisit time, high spatial resolution, broad spatial coverage and multispectral capabilities, which are critical for the river ice monitoring. Techniques developed to identify and map ice cover use a large difference between the reflectance of ice and clear water in the visible and in the near-infrared spectral bands (Satterwhite *et al.*, 2003). Observations in these spectral bands are available from most recent satellite sensors including, in particular, the Moderate Resolution Imaging Spectroradiometer (MODIS) onboard the Terra and Aqua satellites and the Advanced Very High Resolution Radiometer (AVHRR) onboard National Oceanic and Atmospheric Administration (NOAA) and Polar Orbiting Meteorological (MetOp) satellites. Latifovic and Pouliot (2007) identified four lake ice phenology events, which are the beginning and the end of the lake freeze-up and the start and the end of ice breakup using AVHRR near-infrared reflectance. Pavelsky and Smith (2004) combined the visible and near-infrared bands of MODIS and AVHRR sensors for the ice breakup detection in arctic rivers. In another study, Liu and Yan (2005) used the Normalized Difference Snow Index and near-infrared observations from MODIS for ice detection on the Yellow River. MODIS snow and ice products generated by NASA (MOD10_L2 and MYD10_L2) provide daily information on inland snow and sea ice at 500 m spatial resolution (Riggs *et al.*, 2006). However, the land/water mask used in these products is not detailed enough to include small water bodies and rivers, which may be completely or partially missing in the product. Subsequent satellite missions such as the Suomi National Polar Orbiting Partnership/Joint Polar Satellite System and the Geostationary Operational Environmental Satellite-R Series will continue providing moderate and high spatial resolution images in the visible and infrared that can be used for river ice monitoring as well.

Cloud obscuration is the main limitation that hampers the use of optical sensors for the ice cover detection and mapping. Partial remedy to this problem consists in using data from sensors of multiple polar orbiting satellites with different overpass time or from geostationary satellites that provide high-frequency observations and subsequent cloud clear compositing of the imagery on a daily or longer term basis (Gao *et al.*, 2010; Temimi *et al.*, 2011). An example of such approach is offered by Temimi *et al.* (2011), who composited daily observations from Meteosat Second Generation (MSG) SEVIRI instrument to achieve better ice mapping over the Caspian Sea. They demonstrated that the compositing of all daily images from geostationary satellite at 30 min interval results in up to 30% reduction in the cloud-contaminated areas as compared with an instantaneous image.

The objective of this study is to develop an automated technique that uses MODIS data in the visible and near IR spectral bands at 250 m spatial resolution to distinguish between ice-covered and ice-free water on the Susquehanna River, one of the major river in the Northeastern USA. The developed operational system is expected to provide daily maps of the ice cover distribution on the Susquehanna River. The information on ice extent derived from ancillary sources like high-resolution satellite images, aerial photography and observation-based ice charts are used to assess the reliability of the product.

This study was supported by NOAA's National Weather Service, Eastern Region, with the ultimate objective of developing a technique for routine monitoring of river ice extent in major rivers in the USA.

STUDY AREA AND DATA SET

Study area

The Susquehanna River Basin covers 27 510 square miles in portions of the states of New York, Pennsylvania and Maryland. It is one of the largest and longest rivers in the Northeastern USA (Figure 1), with a total length of about 444 miles. The river width is variable and can reach approximately 1 mile (~1609 m) at Harrisburg in the lower subbasin Susquehanna River Basin Commission (SRBC, 2006).

The Susquehanna River Basin is one of the most flood-prone watersheds in the country because of its bathymetry and geomorphology with narrow gorges and low riverbanks. Since the early 1800s, major flood events have been recorded, on average, every 14 years (SRBC, 2006). Compared with other northeastern US rivers, most major flood events along the Susquehanna River were ice related. Damages were substantial during these events because more than 60% of the hydropower capacity in Pennsylvania State is located within the basin (Ice Engineering Group, 1999). The latest major events were observed in 2004 and 2006. According to the data archive of Ice Engineering Group at the CRREL (2011), ice cover may appear on the river as early as the month of November. Depending on meteorological conditions, ice on the Susquehanna may breakup as early as February. However, most often, ice breakup is observed in the month of March (Ice Engineering Group, 1999). Figure 1 shows the location and the number of historical ice jams observed along the entire river since 2002 (CRREL, 2011). Most ice jams events are reported downstream Susquehanna River. It is important to mention that these records may not be comprehensive because they depend on the availability of *in situ* observations for each location (Ice Engineering Group, 1999).

Data set

MODIS instrument onboard the Terra satellite has a viewing swath width of 2330 km that allows it to continuously cover the entire Earth's surface in 1 to 2 days. Observations are made in 36 different spectral bands in the visible, near-infrared and infrared bands. The spatial resolution depends on the spectral band and ranges from 250 m to 1 km. In this study, we have used

Copyright © 2012 John Wiley & Sons, Ltd.

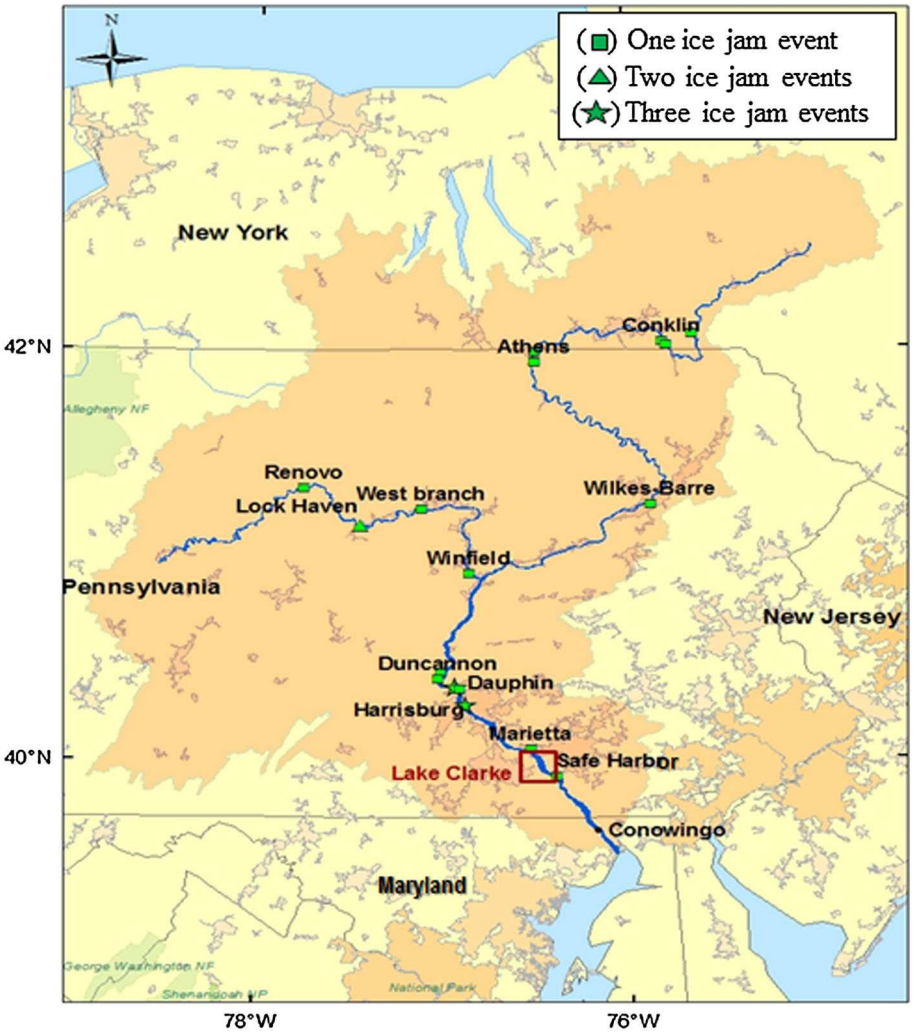


Figure 1. Susquehanna River and watershed location and number of historical ice jams events observed along the river since 2002

daily MODIS-Terra product MOD09GQ, which provides reflectance data in MODIS band 1 (620–670 nm) and in band 2 (841–876 nm) regridded to the sinusoidal projection at the spatial resolution of 250 m. MODIS data for nine winter seasons from 2002 to 2010 have been acquired from the Satellite Receiving Station of NOAA–Cooperative Remote Sensing Science and Technology Center (CREST) at the City College of New York.

Another MODIS product used in the study was the MODIS daily cloud mask at a 1-km spatial resolution (MOD09GA). The cloud mask was resampled to a pixel size of 250 m using the nearest neighbour replication method to match the MODIS reflectance data. The entire watershed of Susquehanna River (within 39°32′N to 42°30'N and 78°31′W to 74°58′W) is completely covered by two granules, h12v04 and h12v05.

Verification and validation of the satellite-based ice mapping results was conducted using several ancillary sources of information on the ice presence on the river. These data sets include high-resolution Landsat 7 ETM+ images, *in situ* river discharge measurements at Conowingo station (39°39′ N, 76°10′W) obtained from the National Water Information System of the USGS observations, aerial photographs and ice bulletins regularly issued by the Safe Harbor Water Power Corporation (SHWPC). The Ice bulletins are based on *in situ* observations and contain reports on the ice coverage on Lake Clarke in the lower section of Susquehanna River.

METHODS

A threshold-based decision tree technique was applied to classify satellite images and to assess the extent of ice coverage (Figure 2). Threshold values were determined using the frequency distribution of the near-infrared reflectance values (Figures 3b and 3c). The reflectance of ice is substantially higher in the visible than that in the shortwave and in the middle infrared. This difference between the spectral signatures of water and ice makes the distinction between them in this spectral domain possible (Comiso and Steffen, 2001). Observations in the near-infrared were selected in this study to differentiate between ice and ice-free water. In the examined MODIS images, lower near-infrared reflectance values correspond

Copyright © 2012 John Wiley & Sons, Ltd.

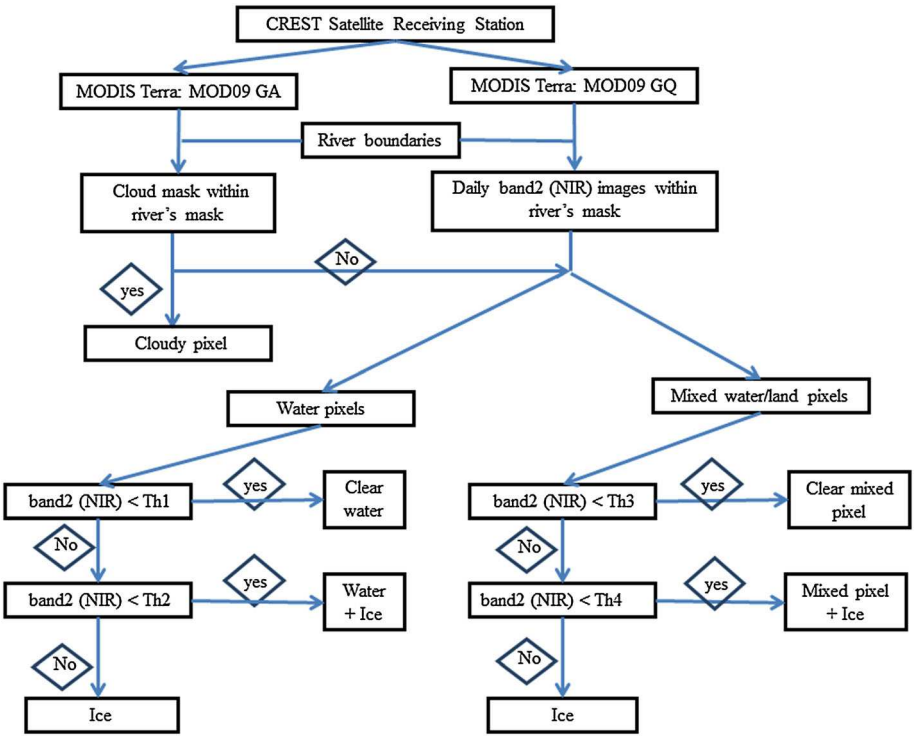


Figure 2. Flowchart of river ice mapping

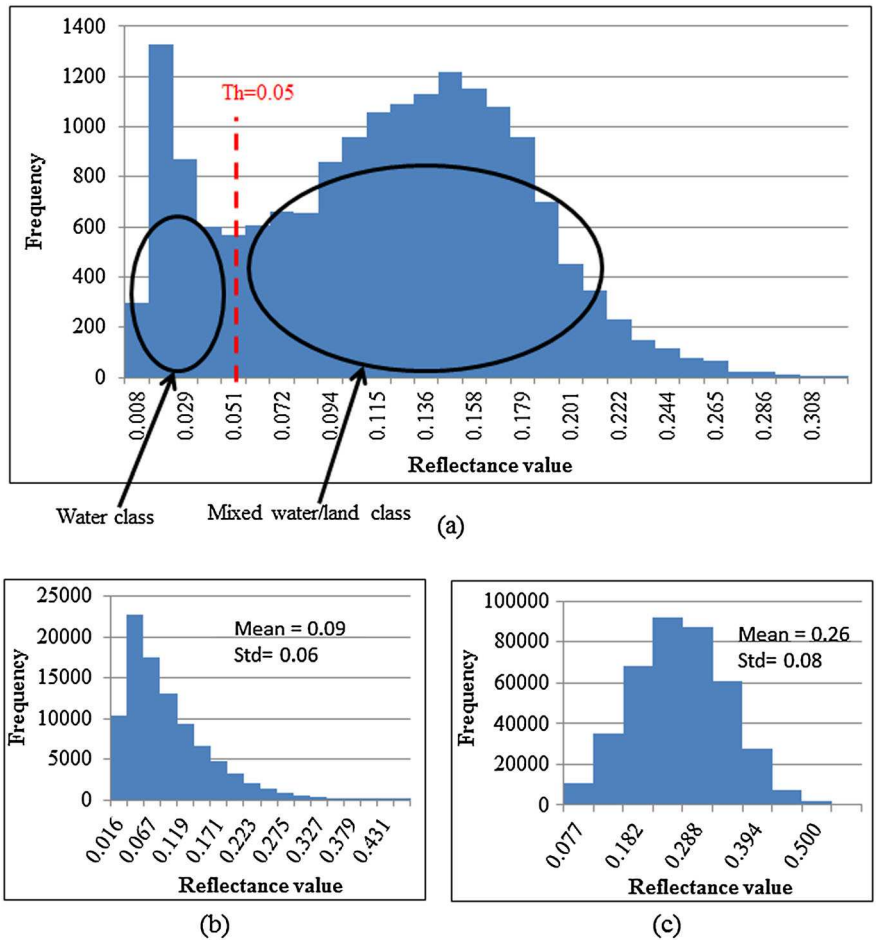


Figure 3. Near-infrared histograms using data (35 images) from summer 2009 (a): Minimum Reflectance using all the images; (b): water class pixels; (c): mixed water/land pixels

Copyright © 2012 John Wiley & Sons, Ltd.

to clear pixels while higher values correspond to ice covered pixels.

River mask generation

The first consideration was given to the delineation of the study area and to the generation of the river's mask, which includes all MODIS grid cells susceptible to be covered by ice in winter time. Also, this first step aims to identify water pixels and mixed pixels that include water and land (i.e. river banks). The delineation of the river banks was first done using a river mask in shape file format created by USGS and provided by the SRBC (2010). The mask was manually refined to include all river pixels particularly in its upstream sections where pixels tend to be a mixture of land and water.

The result of overlaying the river boundaries obtained from USGS database over a MODIS image of the river in the near-infrared band reveals a shift between the river boundaries and MODIS pixels, which can clearly be identified as water. This shift can be attributed to the navigation error of MODIS. The average geolocation error in the MODIS data ranges within 50–150 m at nadir (Wolfe *et al.*, 2002). The river mask was determined using the USGS river limits and refined further through a visual inspection of an image that comprises the minimum value of the near-infrared reflectances of each pixel in 35 MODIS images acquired in summer 2009. Selected summer images were cloud free and exhibited the highest contrast between land and water. Clear land pixels were masked out. The river mask was generated in the sinusoidal projection at the same 250-m spatial resolution as MOD09 products.

The examination of the resulting river mask clearly shows changing river width and the number of MODIS pixels that fall within the river boundaries. Although downstream sections of the river are wide enough to include several MODIS pixels, upstream tributaries are narrower and may comprise one or two MODIS pixels (i.e. 250–500 m width). Because of the difference in the reflectance between pure water and mixed land–water scenes, all pixels in the study area were classified into two categories, namely, 'water' and 'mixed'. The classification was performed by the segmentation of the masked area using the frequency distribution of the pixels reflectance in the near infrared. The latter was derived form 35 summer-time images of MODIS. All 35 summer images were composited and the minimum near-infrared reflectance value for each pixel in the map was retained. Figure 3-a shows the frequency distribution of the near-infrared reflectance in the composite image. According to the frequency distribution in figure 3-a, near-infrared reflectance values below 0.05 correspond to clear water pixels. Mixed pixels, i.e., land and water, have reflectance values above 0.05. The delineation of the study area shows that in the lower Susquehanna River, where the width of the channel exceeds 1 mile (~1609 m), four to five MODIS pixels across the riverbed may be attributed to the water class, while upstream, the river is narrower and most MODIS pixels are mixed land–water scenes (Figure 4).

It is noteworthy that the use of the minimum value of the 35 summer images taken at different dates and therefore different hydraulic condition tend to overestimate the river width and include all possible water pixels even those which may flood occasionally. An overestimation of the river width is tolerable as it enables us to account for the navigation error of the sensor and include all possible MODIS pixels susceptible to be within the river boundaries.

The near-infrared reflectance values obtained from different 35 MODIS scenes show time variability of pixel reflectance within each class, i.e., water class and mixed water and land pixels (Figures 3-b and 3-c). This variability can be explained partially by the change in the geometry of observations, i.e., the view angle and the sun illumination. MODIS data used in this work, MOD09GA and MOD09GQ, are not corrected for the illumination and view angles which determine the land surface directional reflectance. Changes in the viewing angle are more critical when geostationary satellites are in use (Temimi *et al.*, 2011). Since MODIS is a sun-synchronous sensor, which means that it overpasses the same location at the same solar time (i.e., solar illumination), we assume that observations collected during winter season at this latitude should not vary substantially and therefore their impact on ice identification should not be significant. There is a MODIS BRDF (Bidirectional Reflectance Distribution Function) corrected product but it is a 16-day composite which is not appropriate for daily monitoring of ice conditions in the river.

Ice detection

A threshold-based decision tree algorithm was applied to MODIS data to detect ice on the river. The classification algorithm is applied to both pure water and mixed water-land pixels (Figure 2). These two classes are identified hereafter as the river class for water pixels which fall within the river

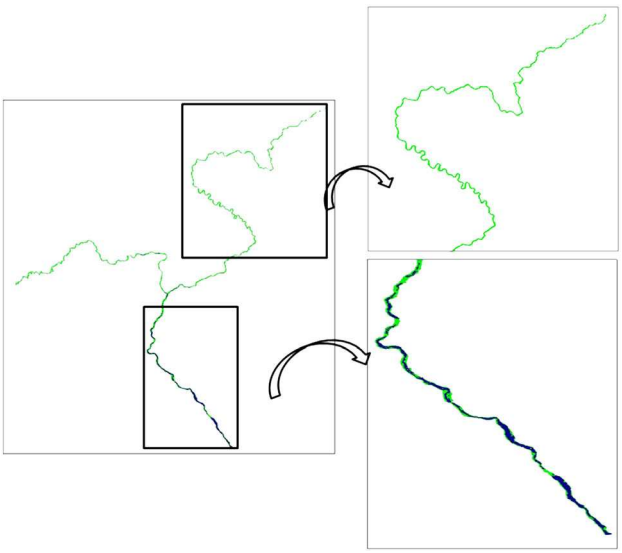


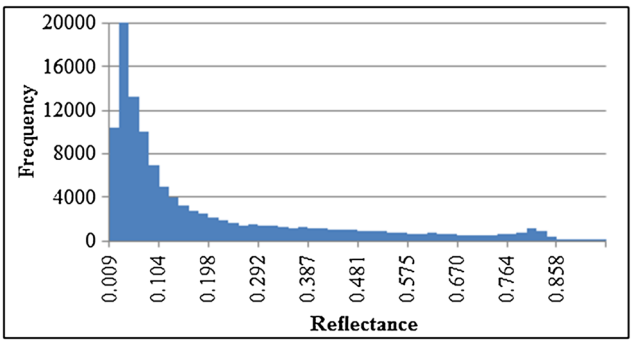
Figure 4. River mask (blue: water pixels, green: mixed water/land pixels)

Copyright © 2012 John Wiley & Sons, Ltd.

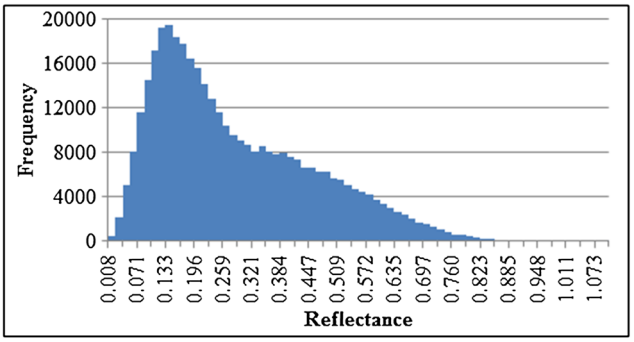
limits and the river-land class for upstream pixels which represent a mixture of land and water.

A set of predetermined threshold values(indicated as Th1, Th2, Th3 and Th4 in Figure 2) is used in the decision tree algorithm to identify water and ice as well as water/ice mixed pixels for each class (i.e., the river class and the mixed river-land class). The threshold values were established empirically through the analysis of the frequency distribution of MODIS near-infrared reflectance over the study area. Figure 5, shows the frequency distribution of the reflectances from 90 images acquired in winter 2004 from January to March. From this figure it is seen that the near-infrared reflectances in winter time were overall larger than those observed in summer (Figure 3). For the "water" class (Figure 5-a), a dominant first peak can be clearly identified and attributed to the "water" class. The histogram is therefore skewed to the left side. Towards the end of the right tail of the histogram, a second peak is seen which corresponds to ice covered pixels within the water class. In the case of mixed land–water pixels (Figure 5-b), the frequency distribution lacks distinct maxima corresponding to different surface classes. This happens since the contribution of different land scenes (including snowy ice-like pixels) may cause the near-infrared reflectance to increase and to reduce the contrast between ice free and ice covered pixels.

For each of the two classes representing river and mixed river-land pixels, two threshold values were selected. The first threshold having the lowest value was selected to identify the "water" scenes. Pixels classified as "water" should have a reflectance lower than the first threshold (Th1). Then, a second threshold (Th2 or Th4) that is higher than the first one was defined to determine ice pixels which have a reflectance value higher than it. Pixels having reflectance that falls between the water threshold and the ice threshold values were considered "mixed", i.e., were assumed to contain both water and ice. According to the frequency distributions determined in figures 3-b and 3-c, the first threshold (Th1 or Th3) which corresponds to the separator of class "water" was selected as the mean of the near-infrared values using 35 summer images acquired during the summer 2009. For the water class, the threshold value of 0.092 (Th1) was selected while for the mixed pixel class, a value of 0.26 (Th3) was considered. The second threshold value, Th2 or Th4, needed to detect pure ice pixels was offset at 0.5 for the river and mixed river-land classes. These threshold values concur within the range of the river ice near-infrared reflectance values determined by Satterwhite *et al.* (2003).



(a) River class pixels



(b) River-Land class pixels

Figure 5. Near-infrared histograms using winter 2004 data (90 images) within each class

The classified daily images were used to generate the weekly and the running composited images to reduce the cloud obscuration and to improve the presentation of the results. The weekly composite product retains all ice covered and cloud free pixels during the week. So, it represents the maximum ice cover observed during the week. In northern locations, where cloudy conditions may persist longer than a week, the weekly composite may not be sufficient to generate a cloud free product. The running composite product is a daily product which comprises the latest cloud free observation for each pixel. The running composite algorithm is initialized with an ice free map which is then updated regularly throughout the winter using daily ice retrievals.

The implemented system automatically acquires MODIS data from the NOAA CREST Satellite Receiving Station Facility as the satellite overflies the New York City area. Then, it generates the daily classified image along with the running and the weekly composite images. An RGB colour composite image is also produced to visually examine the scene and to verify the retrievals. All products are generated at 250 m spatial resolution in the sinusoidal projection. Maps are posted regularly online. A prototype of the automated Web tool is available at http://water.ccny.cuny.edu/crios/.

RESULTS AND DISCUSSION

The examples of daily cloud free RGB colour composite images in figure 6 demonstrate the capability of MODIS images to reveal the presence of ice on Susquehanna River. Since cloudy pixels were masked out, white colour in the RGB composite refers only to ice on the river. In figure 6, black areas correspond to open water while dark green colour refers to mixed pixels consisting of land and water. This figure provides evidence that the spatial resolution of 250 m is appropriate for ice detection and compatible with the extent of the river. These results concur with those obtained by Pavelsky and Smith

Copyright © 2012 John Wiley & Sons, Ltd.

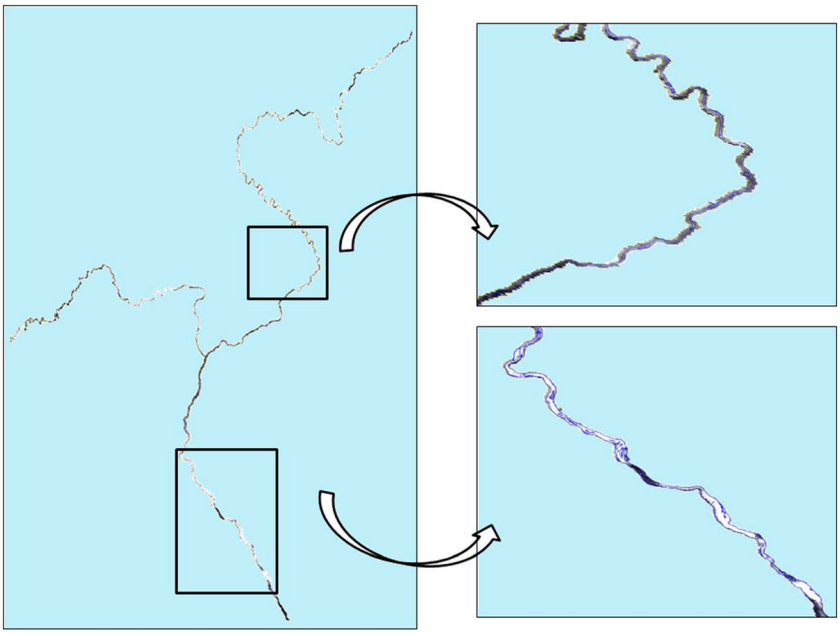


Figure 6. RGB colour compositing images, 02/01/2004 (black colour: clear water; dark green colour: mixed pixel of water and land; white colour: ice and/or snow)

(2004). This figure shows that it is possible to visually identify partially covered pixels as the presence of ice makes the pixels brighter. The goal of subsequent sections is to verify this concept when used in an automated process.

The accuracy of MODIS-based ice mapping results was first assessed using aerial photography taken along the Susquehanna River. Available aerial photographs were taken over the upper and middle branches of the river by the National Weather Service, National Operational Hydrologic Remote Sensing Center, on January 16 2010, between 1:41 PM and 2:03 PM. Each aerial photograph is supplied with information on the location and acquisition time. Since the extent, the projection and the incidence angle of aerial photographs are not provided; they were only used for qualitative comparison with the satellite-based ice product. Figure 7 presents the running composite classified image and the corresponding aerial photographs. The black circle corresponds to the position of the aircraft. Both data sources show mixed water or water/land and ice pixels. Better agreement is especially observed in the pixels with high water fraction (1st aerial photo). For these regions, even low ice concentration can significantly increase the reflectance of a pixel. In contrast, in upstream branches of the river, where most of the observed pixels are a mixture of land and water, the effect of ice on the reflectance is less significant because of the high reflectance of land. This may lead to a misclassification of mixed pixels in the upper Susquehanna. It is noteworthy that low-density"" flowing ice may not be detected particularly in upstream locations since the near-infrared reflectance of shallow or turbid water can be dominant. The consideration of water depth and quality as well as the river bathymetry and geomorphology in the ice detection procedure may improve the classification results. The running composite product, nevertheless, presents a large number of ice covered pixels. This can be explained on one hand by the fact that this product retains the latest clear sky observations which may not adequately represent the current ice condition. On the other hand, it can be explained by possible misclassification errors due to the use of a simple algorithm based on single channel data. Cloud mask errors can also affect the quality of the running composite since missed cloudy pixels may be interpreted as ice.

Besides the aerial photography, *in situ* observations of ice conditions provided by the Safe Harbor Water Power Corp (SHWPC) were used to assess the reliability of the ice detection technique and the accuracy of the river ice product. The comparison of the derived river ice maps to the SHWPC *in situ* observations has demonstrated a good agreement between the two data sets (Figure 8). In fact, the obtained river ice map for December 28, 2010 shows a mixture of water and ice pixels in Lake Clarke. The satellite-based map agrees with the ice cover distribution in chart generated by SHWPC. The shape of the ice covered area on both maps is similar although SHWPC provides more details regarding the type of ice (slush or ice sheet) which are not available from the satellite-based ice maps. On February 1, 2011, different ice conditions were observed in Lake Clarke. The corresponding classified image on February 1st is cloudy so it is compared with the most recent cloud free image obtained on January 31st. The latter presents different portions of clear water and ice covered pixels in addition to mixed pixels of water and ice. Mixed pixels of water and ice result from the relatively coarse MODIS spatial resolution of 250 m. On February 16, 2011, more open water pixels are observed in both scenes which confirms the higher

Copyright © 2012 John Wiley & Sons, Ltd.

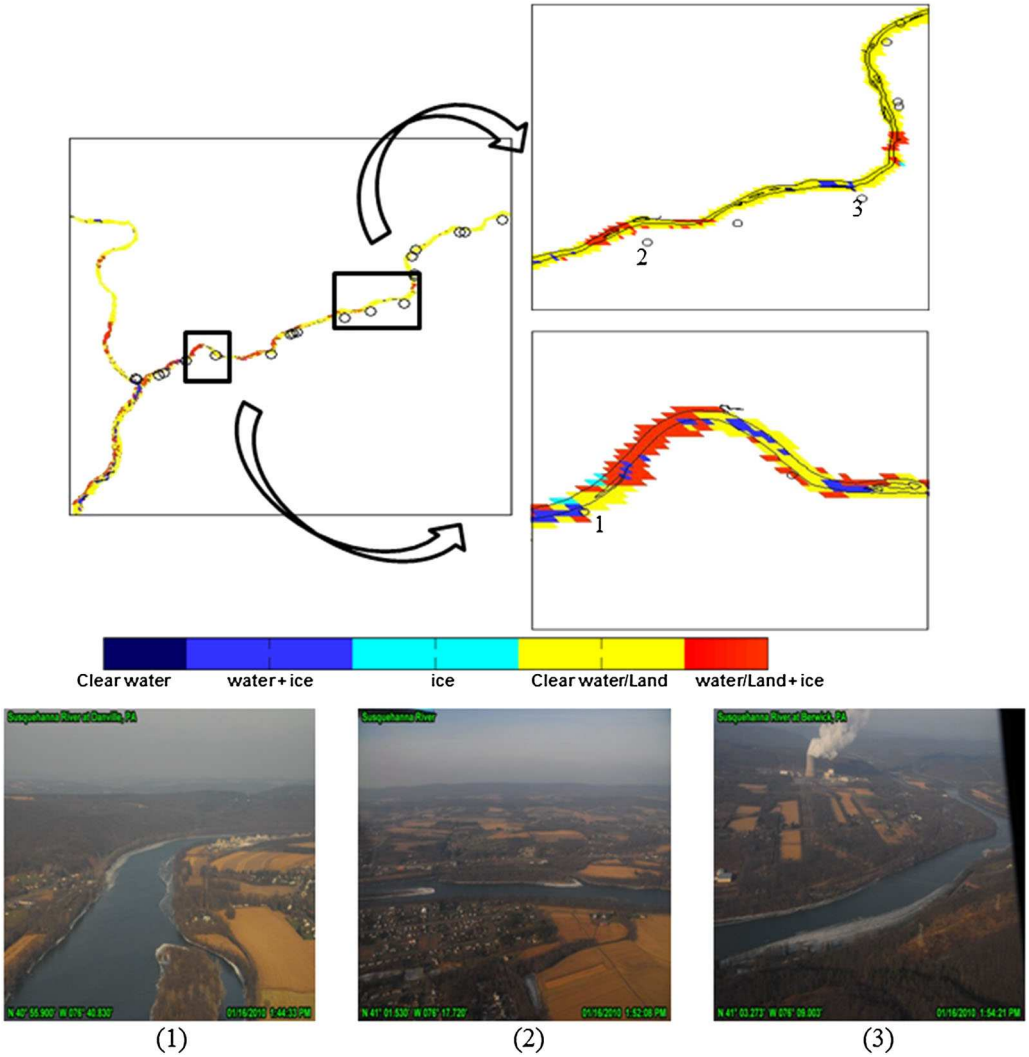


Figure 7. Comparison of the running classified image with the corresponding aerial photography (1, 2, 3 and their corresponding river pixels referred to with circles)

performance of the classification technique to detect water pixels because of the restrictive low threshold value that was selected. So, errors are more likely to occur in the identification of ice pixels rather than the classification of water pixels. For instance, on March 2, 2011, both maps, the ice chart and the generated ice map show that Lake Clarke was ice free.

To further evaluate the agreement between the ice coverage derived from MODIS data and the SHWPC *in situ* observations (considered here as "truth data") in Lake Clarke, we calculated the probability of detection (POD) and the false alarm ratio (FAR) of ice covered pixels (Marzban, 1998; Williams *et al*., 2002):

$$\mathrm{POD} = \mathrm{A}/(\mathrm{A} + \mathrm{C}) \quad (1)$$

$$\mathrm{FAR} = \mathrm{B}/(\mathrm{D} + \mathrm{B}) \quad (2)$$

where: A is the number of pixels of class X which have been correctly classified as class X; B is the number of pixels not of class X which have been identified as X; C is the number of pixels of class X which have not been classified as X; D is the number of pixels not of class X which have been identified as not X; X is the total ice covered pixels class.

Ice charts provided by the SHWPC are not georeferenced and only reflect the extent of ice on the Susquehanna River. As a result, a pixel per pixel comparison with the obtained ice maps was not possible. Therefore, ice maps were compared with SHWPC in two extreme cases when the lake was completely ice covered on December 28, 2010 and when it was completely ice free on March 2, 2011. The goal is to assess the ability of the ice detection system to reproduce these conditions and calculate the corresponding POD and FAR. For this purpose, all MODIS pixels (water or mixed water/land) with ice were considered as fully ice covered pixel. In these conditions, a POD value of 56.1% was obtained with a FAR value of 3% which reflects that around 3% of the pixels were misclassified as ice. Misidentification of ice pixels may be attributed to clouds missed by the MODIS cloud identification algorithm because of their high reflectance in the near infrared.

MODIS ice maps were also compared with a cloud free Landsat 7 ETM + image (Figure 9) at 30 m spatial resolution

Copyright © 2012 John Wiley & Sons, Ltd.

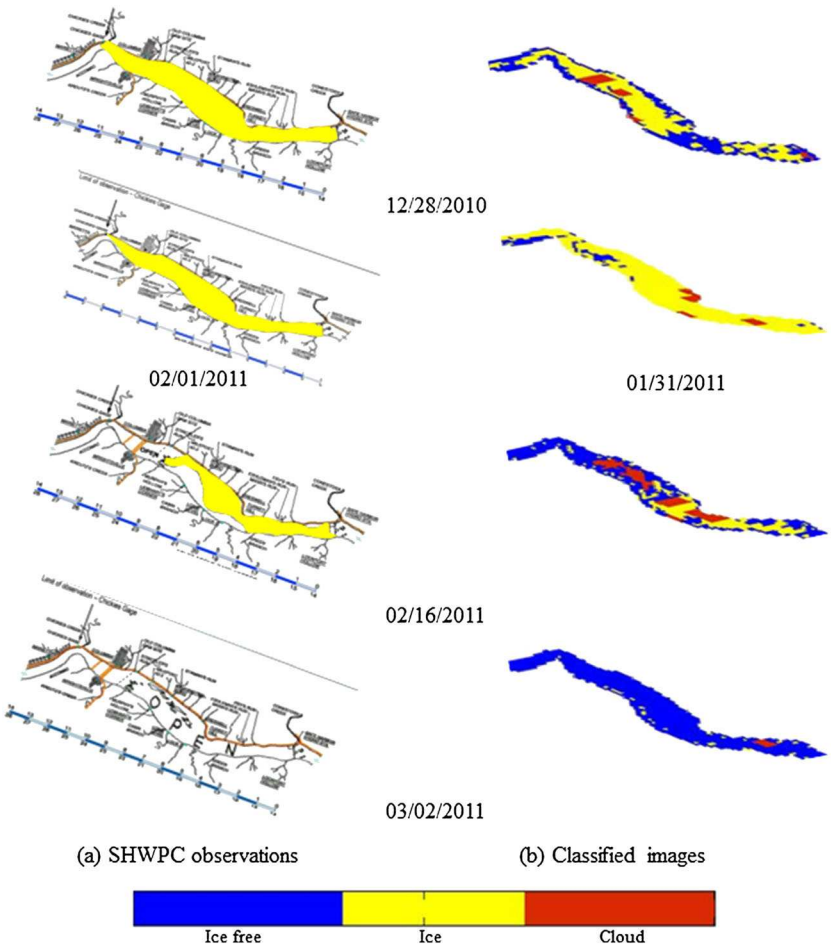


Figure 8. Comparison of the daily classified images and the corresponding SHWPC observations on Lake Clarke

of the lower Susquehanna acquired on February 2, 2005. The Landsat image was first masked using the river boundaries (Figure 9-a). Ice covered areas in the Landsat image were then delineated manually through a visual interpretation of the colour composite (Figure 9-b). The overlay of the determined Landsat ice covered area and the classified MODIS image is presented in figure 9-d. This figure shows a good agreement between the ice coverage derived from MODIS data and the ice coverage derived from Landsat data especially in the lower part of the river. An overestimation of the ice extent was found in the MODIS-based product. The ice fractional coverage of 58.82% is obtained from the MODIS classified image and 43.64% from the Landsat classified map of the lower branch of the river. This overestimation can be attributed to possible misclassification errors especially for mixed river-land sections of the river where snow covered land can be classified as ice covered pixels. In fact, figure 9-c shows high ice coverage concentration along the river boundaries where MODIS pixels are considered as mixed of land and water and where the Landsat image of the entire area surrounding the river reveals the presence of snow. For the POD and FAR calculation, all MODIS pixels with ice were considered as fully ice covered pixels and clear mixed pixels were considered as water pixels. With these assumptions, a POD of 91.2% was obtained with a FAR of 37%. This high FAR value reflects misclassification errors because of the coarse spatial resolution of MODIS data and incorrect interpretation of snow covered lands. It is not straightforward to distinguish between the ice and snow

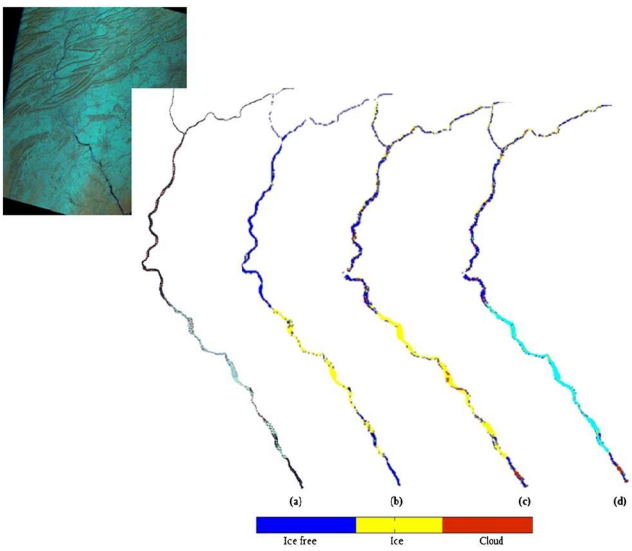


Figure 9. Comparison of the daily classified MODIS images and the corresponding classified Landsat image acquired on 2005/02/02, (a), Landsat RGB image, (b) classified Landsat image, (c) classified MODIS image, (d) overlay of classified Landsat image and MODIS classified image (cyan: ice covered pixels obtained with both images, yellow: ice covered pixels obtained with only MODIS image, blue: clear water)

Copyright © 2012 John Wiley & Sons, Ltd.

covered pixels over mixed land/water pixels. In fact, in early winter the land surface may be covered with snow while the river is ice free. In late winter, the snow cover may melt off earlier than the ice cover in the river. These two cases are hardly distinguishable with one-channel algorithm. Additional observations in other bands in conjunction with *in situ* observations of snow from networks like Snowpack Telemetry (SNOTEL) can be used to refine the classification in these upstream sections of the river.

To assess the consistency of the product further, time series of classified images for winter 2010/2011 (November 2010 through March 2011) of ice in Lake Clarke were extracted (Figure 10). These images show the variability of the spatial distribution of ice extent in Lake Clarke and on adjacent streams. Seasonal change in the ice cover extent can be observed in the images. The beginning of the ice onset occurs in the month of November. Daily analysis of classified images during this month has shown rapid spatial variations of ice cover. This can be attributed to a relatively higher value of the discharge during the month of November because of rainfall events and the capability of the river to rapidly drain all possible ice formations out to Chesapeake Bay. The absence of persistent ice jams along the river fostered the rapid drainage. A larger number of ice pixels were detected in December. This increase can be the result of a decline in the river discharge and of lower air temperature. These two factors foster the formation of ice on the river and reduce its motion and therefore the capability of the river to carry the formed ice sheets to Chesapeake Bay. Heavy ice conditions were dominant during the month of January. Figure 10 shows quasi persistent total ice coverage throughout the month of January. This can be explained either by the fact that they correspond to compact ice sheets or by the reduced drainage during the winter. On January 31st, the river was totally ice covered. Ice breakup occurred in February. In early March (March 2nd image), the river and Lake Clarke were completely ice free.

The climatology of ice extent on Susquehanna River was investigated using MODIS images taken from November 2002 to March 2011. Time series of ice extent were generated using the developed running composite product. The climatology of ice on the Susquehanna was compared with the records of the ice jams obtained from the CRREL Ice Jams Database. Figure 11 shows the interannual variability of the fraction of the river covered by ice. The fraction of ice coverage is the ratio of the total number of pixels identified as ice divided by the total number of river pixels in the MODIS image. Developed times series demonstrated significant year-to-year fluctuations of the ice coverage. In winter of 2004, as well as in late January and in early February 2007, about 50% of the river was ice covered. Less ice, around 30% coverage was observed in February 2003. Figure 11 shows that more ice jams were reported in 2004 and 2007 when the ice concentration in the river was larger than average. Fewer ice jams were reported during years with smaller ice coverage.

The temporal variability of ice extent was compared with the observed discharge as an additional qualitative verification of the product and its consistency. According to the literature, a negative correlation should be expected between the river discharge and ice concentration (Beltaos, 2009; Beltaos and Prowse, 2009). Indeed, ice presence along the river reduces the flow since it retains water upstream and slows down the drainage and consequently reduces the discharge. The temporal variability of the ice extent was compared with the

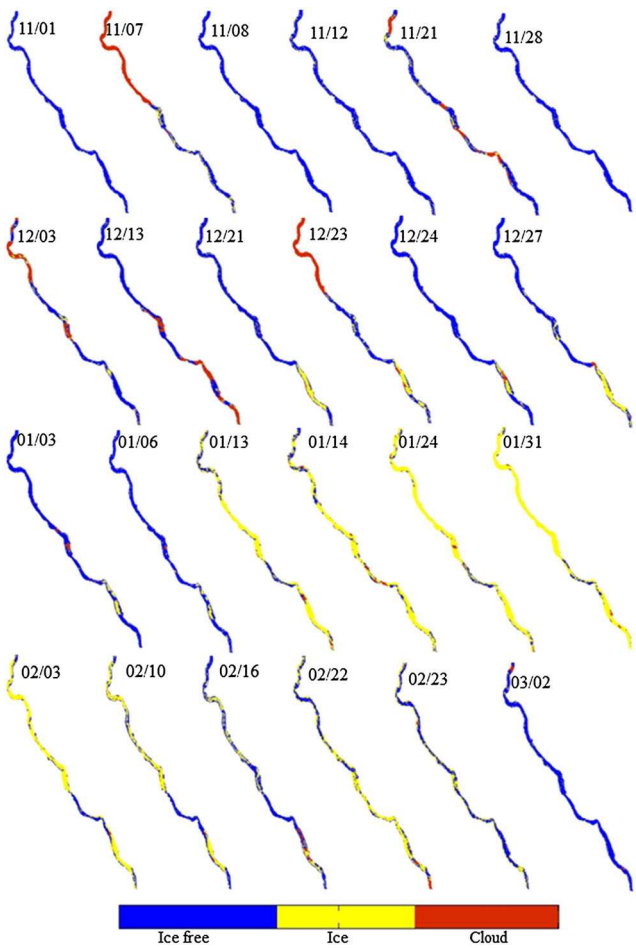


Figure 10. River ice coverage pattern on Lake Clarke from November 2010 to March 2011

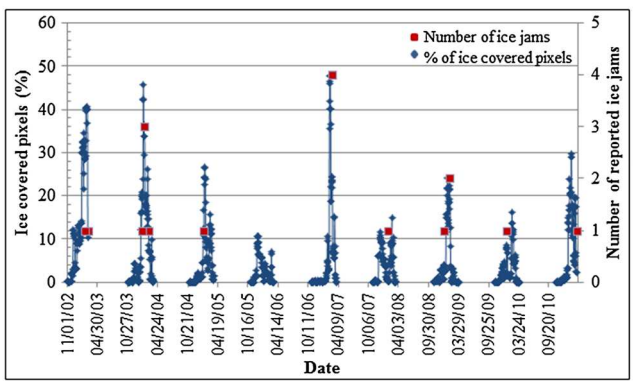


Figure 11. Climatology of the ice coverage and reported ice jams in the Susquehanna River

Copyright © 2012 John Wiley & Sons, Ltd.

observed discharge at Conowingo station (39°39′ N, 76°10′W) which is a downstream station close to the river's outlet (Figure 12). The total ice extent was determined from the total number of pixels classified as ice. In the generated daily ice maps (not shown here), the fluctuation in the number of ice pixels was very significant because of the cloud presence which made the development of a consistent daily time series of ice extent difficult. The weekly composites showed a better correspondence between the extent of river ice and the river discharge. The ice coverage variability is smoother (Figure 12-a). An inverse relationship with observed discharge was noticeable. Figure 12-a clearly shows that during winter time when ice extent increases, the discharge in the river decreases. Peaks of the ice extent coincide with low discharge values. In spring, during the ice melting period, the discharge reaches maximum. High discharge values correspond to lower ice extent. An exponential regression was established with a correlation coefficient of 0.75 (Figure 12-b) between the observed discharge and the ice fraction derived from the weekly composite. The running composite product also reflects a negative correlation between the ice extent and the discharge of 0.55 which is less than the correlation obtained with the weekly composite product (Figure 12-b).

The discharge time series reveal two peaks, one in early January and another in the middle of March. These two peaks coincide with lower ice coverage extent but with a lag time on the order of a few days. The discharge peaks a few days before the ice extent minimum. Ice on the Susquehanna River is typically drained out completely in approximately 3 days (Stephen DiRienzo, NOAA NWS, personal communication). This observed lag can also be attributed to drainage time of groundwater and runoff, particularly from snowmelt. In the middle of March, the discharge reaches a peak because of the increasing snowmelt contribution and warm weather, which fosters the breakup of ice especially in downstream sections of the river while the upper Susquehanna remains ice covered. So, ice tends to disappear from downstream sections before upstream sections since the river cross section downstream is larger and the river capability to drain the ice is better.

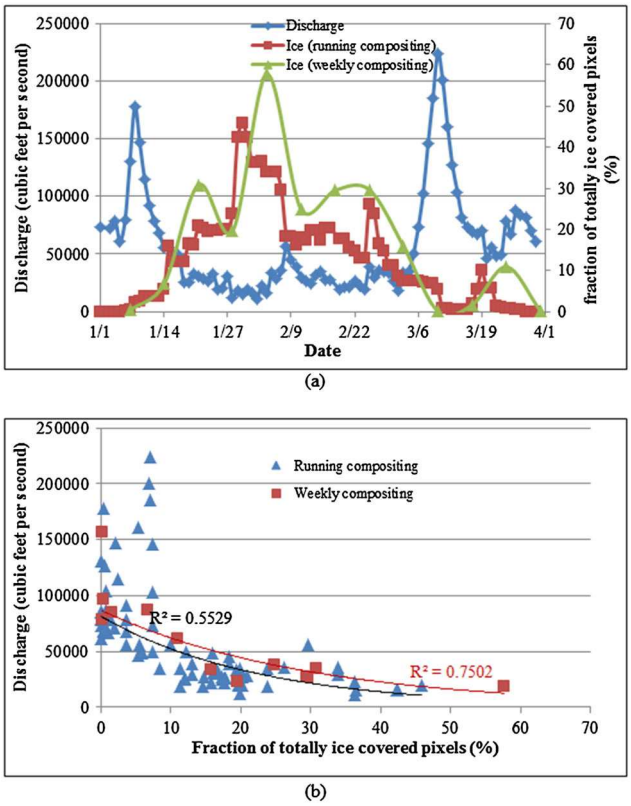


Figure 12. Relationship between ice extent derived from the weekly and running composite products and observed discharge at Conowingo

CONCLUSIONS

MODIS images were used in this study to develop an automated technique to detect and map ice on the Susquehanna River. The performance of the technique is acceptable as the generated maps showed good agreement with time series of discharge and also with *in situ* observations of the ice condition provided by the SHWPC and aerial photos. The ice product was also able to reproduce the intraseasonal cycle of river ice from ice onset to ice breakup and total melting. The performance of the developed system can be enhanced in several ways. First, the cloud mask can be improved. The MODIS global cloud mask currently used in the system is supposed to perform well on a global scale. The development of a local cloud mask which is fine tuned to the local conditions in the Susquehanna River basin may lead to more accurate cloud identification and therefore to reduced FAR. Second, the threshold criteria can be modified to account for changes of the observed reflectance with the geometry of observations. In the current version of the algorithm the adopted thresholds were static and their values did not vary in space and time. Third, the delineation of the river particularly in its upstream portion may be validated using high-resolution topographic maps, for instance, from LIDAR flyovers. In this study, the river mask was generated using summer time MODIS images. The delineation of the study area showed that there are two dominant classes, namely "water" class where the river is large enough to encompass few MODIS pixels and mixed "water-land" class which is mainly in the upper river where the cross section is narrow. The segmentation of the study area into two classes has led to better definition of the threshold values which were used in the ice detection.

This technique can be expanded both in time and space. First, additional years of MODIS images, as well as AVHRR particularly over downstream wide sections of the river may be analyzed and climatology of more than 20 years of ice extent can be generated. The comparison of these long time series to recorded extreme ice related events such as ice jams and ice flood may lead to the development of a relationship between the presence of ice and its spatial distribution along the river and the

Copyright © 2012 John Wiley & Sons, Ltd.

likelihood of such extreme events which may be used then in the development of an early warning system of ice related events. The developed technique can be also transferred to the new National Polar Orbiting Partnership Visible/ Infrared Imager/Radiometer Suite (VIIRS) sensor. Secondly, the proposed approach and the ice mapping system can be easily transferred to other rivers affected by ice. In the northeast, it will be extended to the Hudson and Mohawk Rivers. This will be addressed in future studies.

The developed ice product is posted online and updated on a daily basis under the CREST River Ice Observation System (CRIOS) (http://water.ccny.cuny.edu/crios/) which is a Web tool created to allow end users, NWS forecasters, reservoir managers and the public, to interactively examine ice conditions on the entire river.

ACKNOWLEDGEMENT

The authors would like to acknowledge the contribution of SHWPC through providing ice chart to verify the proposed river ice product. We would also like to thank David Vallee, Stephen DiRienzo and David Radell from NOAA NWS for their comments and suggestions. This study was supported by the NOAA under grant number NA06OAR4810162. The statements contained in this article are not the opinions of the funding agency or government, but reflect the views of the authors.

REFERENCES

Beltaos S. 2009. River flow abstraction due to hydraulic storage at freezeup. *Canadian Journal of Civil Engineering* **36** (3): 519–523.

Beltaos S, Prowse T. 2009. River-ice hydrology in a shrinking cryosphere. *Hydrological Processes* **23**:122–144.

Brakenridge GR, Nghiem SV, Anderson E, Mic R. 2007. Orbital microwave measurement of river discharge and ice status. *Water Resources Research* **43**(4):1–16.

Chokmani K, Ouarda TBMJ, Hamilton S, Ghedira MH, Gingras H. 2008. Comparison of ice-affected streamflow estimates computed using artificial neural networks and multiple regression techniques. *Journal of Hydrology* **349**(3-4): 383–396.

Cold Regions Research and Engineering Laboratory, CRREL. 2011. Ice Jams Database, Bulletins & Surveys. https://rsgis.crrel.usace.army.mil/icejam/

Comiso JC, Steffen K. 2001. Studies of Antartic sea ice concentrations from satellite data and their applications. *Journal of Geophysical Research* **106**: 31362–31385.

Dey B, Moore H, Gregory AF. 1977. Use of satellite imagery for monitoring ice break-up along Mackenzie River, NWT. *Arctic* **30**(4): 234–242.

Gao Y, Xie H, Yao T, Xue C. 2010. Integrated assessment on multi-temporal and multi-sensor combinations for reducing cloud obscuration of MODIS snow cover products of the Pacific Northwest USA. *Remote Sensing of Environment* **114**(8): 1662–1675.

Gherboudj I, Bernier M, Hicks F, Leconte R. 2007. Physical characterization of air inclusions in river ice. *Cold Regions Science and Technology* **49**(3): 179–194.

Hicks F, Beltaos S. 2008. River Ice. *Cold Region Atmospheric and Hydrologic Studies. The Mackenzie GEWEX Experience*, *Hydrological Processes* **2**, Woo MK (ed.). Springer-Verlag: Berlin; 281–305.

Ice Engineering group. 1999. Ice Engineering information exchange bulletins, technical notes. U.S. Army Cold Regions Research and Engineering Laboratory, http://www.crrel.usace.army.mil/ierd/tectran/IERD21.pdf

Latifovic R, Pouliot D. 2007. Analysis of climate change impacts on lake ice phenology in Canada using the historical satellite data record. *Remote Sensing of Environment* **106**(4): 492–507.

Leconte R, Daly S, Gauthier Y, Yankielun N, Berube F, Bernier M. 2009. A controlled experiment to retrieve freshwater ice characteristics from an FM-CW radar system. *Cold Regions Science and Technology* **55**(2): 212–220.

Liu L, Yan J. 2005. The application of MODIS data on ice flood monitoring of yellow river. *Proceedings of SPIE, the International Society for Optical Engineering*, Wuhan, China, SPIE.

Marzban C. 1998. Scalar measures of performance in rare-event situations. *Weather and Forecasting* **13**(2): 753–763.

Pavelsky TM, Smith LC. 2004. Spatial and temporal patterns in Arctic river ice breakup observed with MODIS and AVHRR time series. *Remote Sensing of Environment* **93**(3): 328–338.

Riggs GA, Dorothy KH, Vincent VS. 2006. MODIS Snow products user guide to collection 5. http://modis-snow-ice.gsfc.nasa.gov/

Satterwhite MB, Mitchell H, Hemmer T, Leckie JD. 2003. Field Spectral Signatures of Snow, Ice, and Water. *Proceedings of SPIE, the International Society for Optical Engineering*, Orlando, FL, United states, SPIE.

Seidou O, Ouarda TBMJ, Bilodeau L, Hessami M, St-Hilaire A, Bruneau P. 2006. Modeling ice growth on Canadian lakes using artificial neural networks. *Water Resources Research* **42**(11).

SRBC. 2006. Susquehanna River Basin commission: Suquehanna River Basin. www.srbc.net/pubinfo/docs.

SRBC. 2010. Susquehanna River Basin Commision. www.srbc.net. visited on November 19th 2010.

Temimi M, Leconte R, Brissette F, Chaouch N. 2005. Flood monitoring over the Mackenzie River Basin using passive microwave data. *Remote Sensing of Environment* **98**(2-3): 344–355.

Temimi M, Romanov P, Ghedira H, Khanbilvardi R, Smith K. 2011. Sea ice monitoring over the Caspian Sea using geostationary satellite data. *International Journal of Remote Sensing* **32**(6): 1575–1593

White K, Tuthill AM, Furman L. 2007. Studies of ice jam flooding in the United States. *Extreme Hydrological Events: New Concepts for Security*, Springer Netherlands **78**: 255–268.

Williams RN, Michael KJ, Pendlebury S, Crowther P. 2002. An automated image analysis system for determining sea-ice concentration and cloud cover from AVHRR images of the Antarctic. *International Journal of Remote Sensing* **23**(4): 611–625.

Wolfe RE, Nishihama M, Fleig AJ, Kuyper JA, Roy DP, Storey JC, Patt FS. 2002. Achieving subpixel geolocation accuracy in support of MODIS land science. *Remote Sensing of Environment* **83**(1-2): 31–49.

Copyright © 2012 John Wiley & Sons, Ltd.